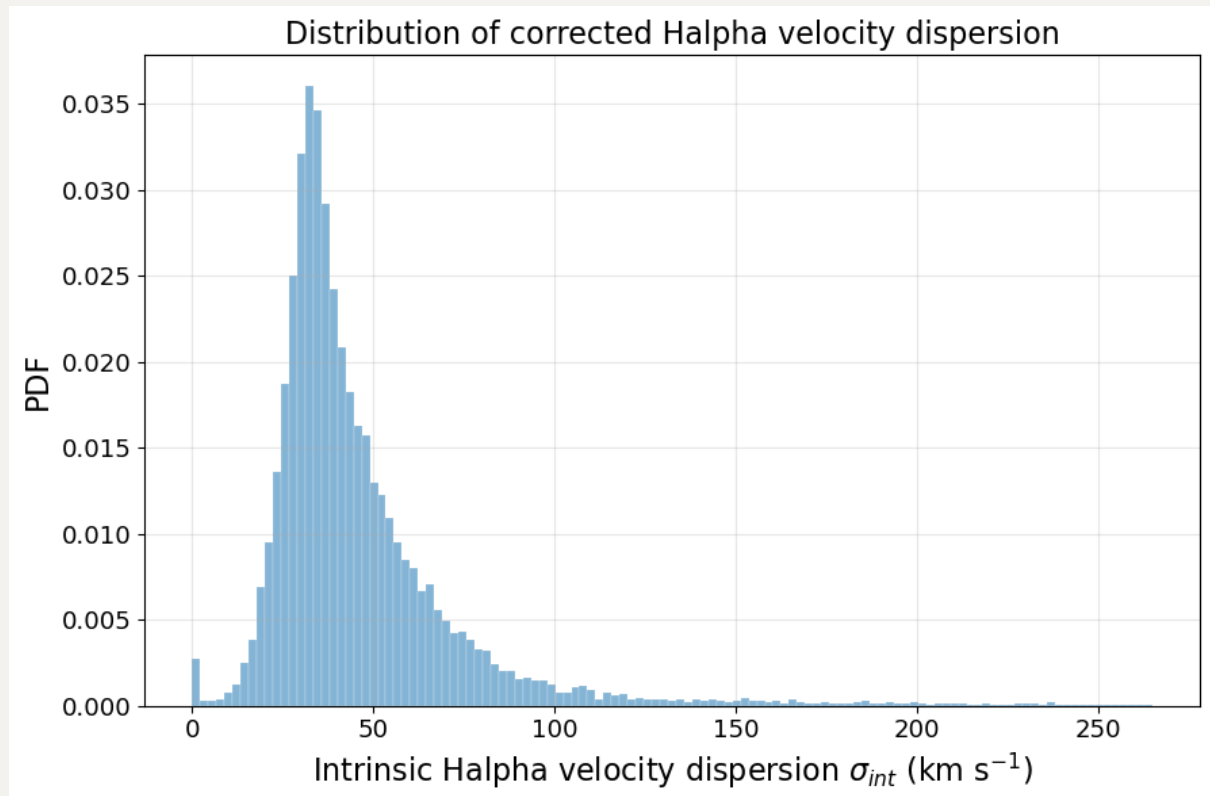


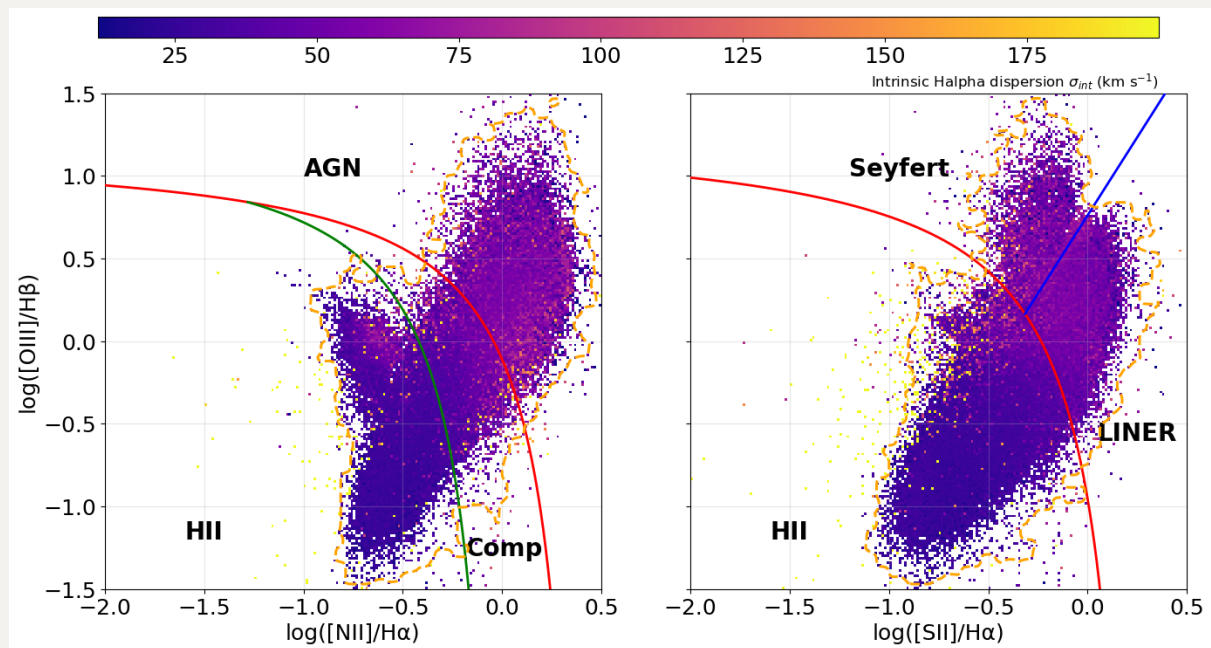
## 20260420 H $\alpha$ dispersion

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Here I subtract the observed H $\alpha$  dispersion by instrumental boardening (48.69 km/s) in quadrature to obtain the intrinsic dispersion. Below I show the distrubution of intrinsic dispersion first.



Then i colorcode the BPT diagram with the intrinsic dispersion. So again, only a few spaxels are rejected in the selection.



Finally, I show the table that report the numbers and fractions of our selection criteria in each galaxy and combined case. In total, HII gives 31.61%, HII+EW gives 29.89% and HII+EW+dispersion gives 29.28%.

Selection chain (per galaxy):

Base(Cell1) -> HII -> HII+EW>6 -> HII+EW>6+sigma\_int<77.5

| Galaxy    | Base(C1)        | HII      | f(HII) | HII+EW>6 |
|-----------|-----------------|----------|--------|----------|
| f(HII+EW) | HII+EW+sig<77.5 | f(final) |        |          |
| IC3392    | 49984           | 14906    | 0.2982 | 13877    |
| 0.2776    | 13811           | 0.2763   |        |          |
| NGC4064   | 111655          | 6946     | 0.0622 | 6343     |
| 0.0568    | 6172            | 0.0553   |        |          |
| NGC4192   | 458926          | 146165   | 0.3185 | 139526   |
| 0.3040    | 139472          | 0.3039   |        |          |
| NGC4293   | 141156          | 4222     | 0.0299 | 517      |
| 0.0037    | 502             | 0.0036   |        |          |
| NGC4298   | 277437          | 156571   | 0.5643 | 143910   |
| 0.5187    | 143910          | 0.5187   |        |          |
| NGC4330   | 88572           | 42589    | 0.4808 | 38371    |
| 0.4332    | 37672           | 0.4253   |        |          |
| NGC4383   | 127766          | 75134    | 0.5881 | 74737    |
| 0.5850    | 59747           | 0.4676   |        |          |
| NGC4396   | 155963          | 81729    | 0.5240 | 81493    |
| 0.5225    | 81470           | 0.5224   |        |          |

|         |         |        |        |        |
|---------|---------|--------|--------|--------|
| NGC4419 | 126953  | 7288   | 0.0574 | 5810   |
| 0.0458  | 5588    | 0.0440 |        |        |
| NGC4457 | 227739  | 6280   | 0.0276 | 6050   |
| 0.0266  | 5798    | 0.0255 |        |        |
| NGC4501 | 578518  | 261414 | 0.4519 | 251053 |
| 0.4340  | 251045  | 0.4339 |        |        |
| NGC4522 | 82367   | 39285  | 0.4770 | 38194  |
| 0.4637  | 38112   | 0.4627 |        |        |
| NGC4694 | 59616   | 2584   | 0.0433 | 1935   |
| 0.0325  | 1935    | 0.0325 |        |        |
| NGC4698 | 218649  | 10141  | 0.0464 | 6891   |
| 0.0315  | 6869    | 0.0314 |        |        |
| -----   |         |        |        |        |
| -----   |         |        |        |        |
| ALL     | 2705301 | 855254 | 0.3161 | 808707 |
| 0.2989  | 792103  | 0.2928 |        |        |